

Using Sub queries to Solve Queries

1.Create a report that displays EMPNO, ENAME, DEPTNO of employees who work with that employee. (Nested Sub Query)

Query:

```
SELECT employee_id, first_name, department_id
FROM employees
WHERE job_id=(SELECT job_id FROM employees WHERE first_name='Steven');
```

Output:

```
SQL> SELECT employee_id, first_name, department_id
  2  FROM employees
  3  WHERE job_id=(SELECT job_id FROM employees WHERE first_name='Steven');

EMPLOYEE_ID FIRST_NAME                                DEPARTMENT_ID
-----
      100 Steven                                      90

SQL>
```

2.Display the employees who earn more than the average salary of EMP table. (Nested Sub Query)

Query:

```
SELECT employee_id,first_name,salary
FROM employees
WHERE salary>ALL(SELECT AVG(salary) FROM employees GROUP BY department_id);
```

Output:

```

EMPNO ENAME      SAL
-----
 7566 JONES      2975
 7698 BLAKE      2850
 7782 CLARK      2450
 7788 SCOTT      3000
 7839 KING      5000
 7902 FORD      3000

6 rows selected.
```

3.Display the ENAME, JOB who are Managers. (EXISTS Operator)

Query:

```
SELECT employee_id,first_name,job_id
FROM employees
WHERE employee_id IN (SELECT manager_id FROM departments);
```

Output:

```
SQL> SELECT employee_id,first_name,job_id
  2  FROM employees
  3  WHERE employee_id IN (SELECT manager_id FROM departments);

EMPLOYEE_ID FIRST_NAME
-----
JOB_ID
-----
      200 Jennifer
AD_ASST

      201 Michael
MK_MAN

      114 Den
PU_MAN

EMPLOYEE_ID FIRST_NAME
-----
JOB_ID
-----
      203 Susan
HR_REP

      121 Adam
ST_MAN

      103 Alexander
IT_PROG

EMPLOYEE_ID FIRST_NAME
-----
JOB_ID
-----
      100 Steven
AD_PRES

7 rows selected.

SQL> |
```

4.Display the employees who earn less than the least salary of DEPTNO 10. (ALL Operator)

Query:

```
SELECT first_name,last_name,salary,department_id
FROM employees
WHERE salary=(SELECT MIN(salary) FROM employees WHERE department_id=10);
```

Output:

```
SQL> SELECT first_name,last_name,salary,department_id
  2  FROM employees
  3  WHERE salary=(SELECT MIN(salary) FROM employees WHERE department_id=10);

FIRST_NAME
-----
LAST_NAME          SALARY DEPARTMENT_ID
-----
Jennifer
Whalen              4400      10

SQL> |
```

5. Display the employees who have the same DEPTNO and MGR of a given employee, excluding that employee. (Nested Sub Query)

Query:

```
SELECT empno, ename, deptno, mgr
FROM emp
WHERE (deptno, mgr)=(SELECT deptno, mgr
FROM emp
WHERE ename='Ward'
)AND ename<>'Ward';
```

Output:

```
SQL> SELECT empno, ename, deptno, mgr
  2  FROM emp
  3  WHERE (deptno, mgr) =
  4  (
  5      SELECT deptno, mgr
  6      FROM emp
  7      WHERE ename = 'WARD'
  8  )
  9  AND ename <> 'WARD';

EMPNO ENAME      DEPTNO  MGR
-----
7499 ALLEN        30      7698
7654 MARTIN      30      7698
7844 TURNER      30      7698
7900 JAMES       30      7698

SQL> |
```

6. Write a query that displays the employee number and name of all employees who work in a department with any employee whose name contains a R.

Query:

```
SELECT employee_id,first_name||' '||last_name AS name
FROM employees
WHERE department_id IN (SELECT department_id FROM employees WHERE LOWER(first_name||last_name)
LIKE '%r%');
```

Output:

```
SQL> SELECT employee_id,first_name||' '||last_name AS name
2 FROM employees
3 WHERE department_id IN (SELECT department_id FROM employees WHERE LOWER(first_name||last_name) LIKE '%r%');
```

EMPLOYEE_ID

NAME

100
Steven King

101
Neena Kochhar

102
Lex De Haan

EMPLOYEE_ID

NAME

103
Alexander Hunold

104
Bruce Ernst

105
David Austin

EMPLOYEE_ID

NAME

106
Valli Pataballa

107
Diana Lorentz

108
Nancy Greenberg

EMPLOYEE_ID

NAME

109
Daniel Faviet

110
John Chen

111
Ismael Sciarra

EMPLOYEE_ID

7. The HR department needs a report that displays the ename, deptno, job of all employees whose work in NEW YORK.(Nest Sub Query)

Query:

```
SELECT employee_id,first_name,department_id
FROM employees
WHERE department_id IN (SELECT department_id FROM departments WHERE location_id=(SELECT
location_id FROM locations WHERE city='South San Francisco'));
```

Output:

```
SQL> SELECT employee_id,first_name,department_id
2  FROM employees
3  WHERE department_id IN (SELECT department_id FROM departments WHERE location_id=(SELECT location_id FROM locations WHERE city='South San Francisco'));

EMPLOYEE_ID FIRST_NAME DEPARTMENT_ID
-----
120 Matthew      50
121 Adam         50
122 Payam        50
123 Shanta       50
124 Kevin        50
176 Jonathon     50
177 Jack         50
178 Kimberely    50

8 rows selected.

SQL> |
```

8. Modify the query so that the user is prompted for a LOC.

Query:

```
SELECT ENAME, DEPTNO, JOB FROM EMP WHERE DEPTNO=( SELECT DEPTNO FROM DEPT
WHERE LOC='&LOC' );
```

Output:

```
SQL> SELECT ename, deptno, job
2  FROM emp
3  WHERE deptno =
4  (
5      SELECT deptno
6      FROM dept
7      WHERE loc='&LOC'
8  );
Enter value for loc: Dallas
old 7:  WHERE loc='&LOC'
new 7:  WHERE loc='Dallas'

no rows selected

SQL> |
```

9. Create a report for HR that displays the name and salary of every employee who reports to King.(Nest Sub Query)

Query:

```
SELECT first_name,salary
FROM employees
WHERE manager_id=(SELECT employee_id FROM employees WHERE last_name='King');
```

Output:

```
SQL> SELECT first_name,salary
2 FROM employees
3 WHERE manager_id=(SELECT employee_id FROM employees WHERE last_name='King');

FIRST_NAME          SALARY
-----
Neena                17000
Lex                  17000
Den                  11000
Matthew              8000
Adam                 8200
Payam                7900
Shanta               6500
Kevin                5800
Michael              13000

9 rows selected.

SQL> |
```

10. Write a query to display all the employees working with JAMES

Query:

```
SELECT * FROM employees
WHERE department_id=(SELECT department_id FROM employees WHERE first_name='James');
```

Output:

```
SQL> SELECT * FROM employees
2 WHERE department_id=(SELECT department_id FROM employees WHERE first_name='James');

EMPLOYEE_ID FIRST_NAME
-----
LAST_NAME
-----
EMAIL          PHONE_NUMBER
-----
HIRE_DATE JOB_ID          SALARY MANAGER_ID DEPARTMENT_ID
-----
Mavris
SMAVRIS
07-JUN-02 HR_REP          6500      101      40

EMPLOYEE_ID FIRST_NAME
-----
LAST_NAME
-----
EMAIL          PHONE_NUMBER
-----
HIRE_DATE JOB_ID          SALARY MANAGER_ID DEPARTMENT_ID
-----
Smith
JSMITH
21-SEP-05 HR_REP          6000      203      40

SQL> |
```

11. Display all the employees who earn less than the average salary of their respective departments. (Correlated Sub Query)

Query:

```
SELECT first_name,salary,department_id
FROM employees e
WHERE salary<(SELECT AVG(salary) FROM employees WHERE department_id=e.department_id);
```

Output:

```

SQL> SELECT first_name,salary,department_id
2 FROM employees e
3 WHERE salary<(SELECT AVG(salary) FROM employees WHERE department_id=e.department_id);

```

FIRST_NAME	SALARY	DEPARTMENT_ID
Neena	17000	90
Lex	17000	90
David	4800	60
Valli	4800	60
Diana	4200	60
Daniel	9000	20
John	8200	20
Ismail	7700	20
Jose	7800	20
Luis	6900	20
Alexander	3100	30

FIRST_NAME	SALARY	DEPARTMENT_ID
Shelli	2900	30
Sigal	2800	30
Guy	2600	30
Karen	2500	30
Shanta	6500	50
Kevin	5800	50
Kimberely	7000	50
James	6000	40

19 rows selected.

```

SQL> |

```

12. Write a query to display the LOC and average salary of Each location.(Scalar Sub query).

Query:

```

SELECT d.loc,(
    SELECT AVG(e.sal)
    FROM emp e
    WHERE e.deptno=d.deptno) AS avg_salary FROM dept d;

```

Output:

```

SQL> SELECT d.loc,
2 (
3     SELECT AVG(e.sal)
4     FROM emp e
5     WHERE e.deptno=d.deptno
6 ) AS avg_salary
7 FROM dept d;

```

LOC	AVG_SALARY
DALLAS	2175
CHICAGO	1566.66667
NEW YORK	2916.66667
BOSTON	

```

SQL> |

```

13 Write a query to display the least N salaries. Use In-Line Views.Query:

```

SELECT * FROM (SELECT employee_id,first_name,salary FROM employees ORDER BY salary) WHERE
ROWNUM<=5;

```

Output:

```
SQL> SELECT * FROM (SELECT employee_id,first_name,salary FROM employees ORDER BY salary) WHERE ROWNUM<=5;
```

EMPLOYEE_ID	FIRST_NAME	SALARY
119	Karen	2500
118	Guy	2600
117	Sigal	2800
116	Shelli	2900
115	Alexander	3100

```
SQL> |
```

14. Display the last N rows from the EMP table. (Correlated Sub Query)

Query:

```
SELECT * FROM (SELECT * FROM employees ORDER BY employee_id DESC) WHERE ROWNUM<=5;
```

Output:

```
SQL> SELECT * FROM (SELECT * FROM employees ORDER BY employee_id DESC) WHERE ROWNUM<=5;
```

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	MANAGER_ID	DEPARTMENT_ID
284	James	Smith	JSMITH	111134	21-SEP-05	HR_REP	6000	203	40
283	Susan	Mavris	SMAVRIS	111133	07-JUN-02	HR_REP	6500	101	40
201	Michael	Hartstein	MHARTSTE	111132	17-FEB-04	PM_MAN	13000	100	20
200	Jennifer	Whalen	JWHALEN	111131	17-SEP-03	AD_ASST	4400	101	10
178	Kimberely	Grant	KGRANT	111130	24-MAY-07	SA_REP	7000	120	50

15. Display the employees and sort only employees working in DALLAS. (Scalar Sub Query)

Query:

```
SELECT * FROM employees WHERE department_id IN (SELECT department_id FROM departments WHERE location_id=(SELECT location_id FROM locations WHERE city='Dallas')) ORDER BY salary;
```

Output:

```
12 rows selected.

SQL> SELECT * FROM employees WHERE department_id IN (SELECT department_id FROM departments WHERE location_id=(SELECT location_id FROM locations WHERE city='Dallas')) ORDER BY salary;

EMPLOYEE_ID FIRST_NAME
-----
LAST_NAME
-----
EMAIL
-----
PHONE_NUMBER
-----
HIRE_DATE JOB_ID SALARY MANAGER_ID DEPARTMENT_ID
-----
107 Diana
Lorentz
DLORENTZ
07-FEB-07 IT_PROG 4200 103 60

EMPLOYEE_ID FIRST_NAME
-----
LAST_NAME
-----
EMAIL
-----
PHONE_NUMBER
-----
HIRE_DATE JOB_ID SALARY MANAGER_ID DEPARTMENT_ID
-----
108 David
Austin
DAUSTIN
25-JUN-05 IT_PROG 4800 103 60

EMPLOYEE_ID FIRST_NAME
-----
LAST_NAME
-----
EMAIL
-----
PHONE_NUMBER
-----
HIRE_DATE JOB_ID SALARY MANAGER_ID DEPARTMENT_ID
-----
106 Valli
Pataballa
VPATABAL
05-FEB-06 IT_PROG 4800 103 60

EMPLOYEE_ID FIRST_NAME
-----
LAST_NAME
-----
EMAIL
-----
PHONE_NUMBER
-----
HIRE_DATE JOB_ID SALARY MANAGER_ID DEPARTMENT_ID
-----
104 Bruce
Ernst
BERNST
21-MAY-07 IT_PROG 6000 103 60

EMPLOYEE_ID FIRST_NAME
-----
LAST_NAME
-----
EMAIL
-----
PHONE_NUMBER
-----
HIRE_DATE JOB_ID SALARY MANAGER_ID DEPARTMENT_ID
-----
113 Luis
Popp
LPOPP
07-DEC-07 FI_ACCOUNT 6900 108 20

EMPLOYEE_ID FIRST_NAME
-----
LAST_NAME
-----
EMAIL
-----
PHONE_NUMBER
-----
HIRE_DATE JOB_ID SALARY MANAGER_ID DEPARTMENT_ID
-----
111 Ismael
Sciarra
ISCARRA
30-SEP-05 FI_ACCOUNT 7700 108 20

EMPLOYEE_ID FIRST_NAME
-----
LAST_NAME
-----
EMAIL
-----
PHONE_NUMBER
-----
HIRE_DATE JOB_ID SALARY MANAGER_ID DEPARTMENT_ID
-----
112 Jose
Manuel
JMANUEL
07-MAR-06 FI_ACCOUNT 7800 108 20
```

16. Display the employees who earn a salary less than the average salary of their respective department. Also display the average salary. (Inline View)

```
12 rows selected.
SQL> SELECT e.empno,
2      e.ename,
3      e.sal,
4      v.avg_sal
5 FROM emp e
6 JOIN
7 (
8   SELECT deptno,
9         AVG(sal) avg_sal
10  FROM emp
11  GROUP BY deptno
12 ) v
13 ON e.deptno=v.deptno
14 WHERE e.sal<v.avg_sal;

EMPNO ENAME      SAL      AVG_SAL
-----
7369 SMITH        800       2175
7876 ADAMS       1100       2175
7521 WARD       1250 1566.66667
7654 MARTIN     1250 1566.66667
7844 TURNER     1500 1566.66667
7900 JAMES        950 1566.66667
7782 CLARK      2450 2916.66667
7934 MILLER     1300 2916.66667

8 rows selected.
SQL>
```

17. Write a query that display the LOC of those DEPTS that have a sum of sal less than the avgsal of all the employees in DEPT table. (Use WITH Clause).

Query:

```
WITH dept_sal AS (
SELECT d.location_id,SUM(e.salary) total_sal FROM departments d JOIN employees e ON
d.department_id=e.department_id GROUP BY d.location_id)
SELECT * FROM dept_sal;
```

Output:

```
SQL> WITH dept_sal AS (
2  SELECT d.location_id,SUM(e.salary) total_sal FROM departments d JOIN employees e ON d.department_id=e.department_id GROUP BY d.location_id)
3  SELECT * FROM dept_sal;

LOCATION_ID TOTAL_SAL
-----
1100      74900
1200      93400
1300      24900
1000      60400

SQL> |
```

